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24 March 2026

**5.5 Classwork: Cumulative Review**

1. A particle moves along a straight line with velocity $v(t) = 3t^2 - 12t + 9 \text{ m s}^{-1}$, for $t \geq 0$.

   (a) Find the times when the particle is at rest. [3]

   (b) Find an expression for the acceleration $a(t)$. [2]

   (c) Find the acceleration when $t = 1$. State whether the particle is speeding up or slowing down at this instant. Justify your answer. [3]

   (d) Find the total distance travelled by the particle in the first 3 seconds. [5]

2. In a class of 30 students, 20 study Biology and 18 study Chemistry. Every student studies at least one of these subjects.

   (a) Find the number of students who study both Biology and Chemistry. [2]

   A student is chosen at random from the class. Let $B$ be the event "the student studies Biology" and $C$ be the event "the student studies Chemistry."

   (b) Find $\mathrm{P}(B \cap C)$. [1]

   (c) Find $\mathrm{P}(B \mid C)$. [2]

   (d) Determine whether the events $B$ and $C$ are independent. Justify your answer. [3]

   It is known that 75% of the Biology students and 50% of the Chemistry-only students passed their most recent exam.

   (e) A student is chosen at random and is found to have passed. Find the probability that this student studies Biology. [4]

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3. A geometric sequence has the first term $u_1 = 5$ and common ratio $r = 1.1$.

   (a) Find the 20th term of the geometric sequence. [2]

   (b) Find the sum of the first 20 terms of the geometric sequence. [2]

   (c) After how many terms does the geometric sequence first exceed 50? Show your working. [3]

4. A student records the number of hours spent studying ($x$) and the test score ($y$) for eight classmates. The results are shown below.

| $x$ (hours) | 1.5 | 2.0 | 3.0 | 3.5 | 4.0 | 5.0 | 5.5 | 6.0 |
|---|---|---|---|---|---|---|---|---|
| $y$ (score) | 42 | 48 | 56 | 55 | 65 | 68 | 74 | 78 |

(a) Use your GDC to find the equation of the regression line $y$ on $x$. Write the equation in the form $y = mx + c$, giving $m$ and $c$ to three significant figures. [3]

(b) Write down the value of the Pearson correlation coefficient, $r$. [1]

(c) Describe the correlation between hours studied and test score. [1]

(d) Use your regression line to estimate the test score for a student who studies for 4.5 hours. [2]

(e) Explain why it would not be appropriate to use this model to predict the score for a student who studies for 15 hours. [1]